

Bachelor in Telematics Engineering
2024-2025

Bachelor Thesis

“Machine Learning-Based Predictive Modeling of Energy Prices”

Rodrigo De Lama Fernández

Emilio Parrado Hernández

Leganés, Spain, June 16th 2025



This work is licensed under Creative Commons **Attribution – Non Commercial – Non Derivatives**

SUMMARY

This project focuses on the development of a machine learning-based predictive model for electricity prices in Spain. Using historical data from the Iberian Energy Market Operator (OMIE) and technical analysis indicators, the model aims to accurately forecast hourly energy prices. The study focuses on a single hourly slot, evaluating the performance of various machine learning models, such as Linear Regression, Lasso, and Random Forest. The project focuses on using a sliding window approach alongside feature engineering, employing technical analysis by calculating moving averages, momentum indicators, and volatility indicators. The results display the impact of using technical indicators as features to improve model accuracy and offer insights into the potential of data-driven approaches for energy price forecasting.

Keywords:

Energy, Machine Learning, Sliding Window, Technical Analysis (TA), Technical Indicators.

DEDICATION AND ACKNOWLEDGMENTS

Dedicated to my late grandfather, who was not able to see me become an engineer like himself. To my parents, that have always helped me push through hard moments. To my grandparents who will proudly see me become an engineer. To my amazing girlfriend that has supported me though all the ups and downs of life. To my all of friends and colleagues that have accompanied me these years. To anyone and everyone that has supported me during my years in university. Here is to the next steps in life.

I would like to acknowledge, and thank everyone that has helped me and supported me during the development of this project, highlighting the invaluable support of my tutor, my parents and my girlfriend. I would also like to extend a special thank you to all of the open source contributors of the libraries that have made this project possible.

CONTENTS

1. INTRODUCTION TO THE PROBLEM	1
2. BACKGROUND AND RELATED WORK.	2
2.1. Training a Non-Stationary Dataset with a Sliding Window Approach	3
2.2. Machine Learning Models Selection	3
2.3. Model Optimization: Hyperparameter Tuning and Feature Engineering	4
2.4. Evaluation Metrics	6
2.5. Related Work	7
3. PROPOSAL: SYSTEM DESCRIPTION	8
3.1. Prior Set Up and Configuration	9
3.2. Data Preparation: Ingestion and Cleanup	9
3.3. Feature Matrix Creation	11
3.4. Machine Learning Model Training	13
4. EXPERIMENTATION	16
4.1. Data Description	16
4.2. Model Set Up Explanation	18
4.3. Results of the Testing	19
4.4. Discussion	19
5. REGULATORY FRAMEWORK	21
5.1. Data Availability	21
5.2. Software & Licenses	21
6. SOCIO-ECONOMIC ENVIRONMENT	24
6.1. Impact	24
6.2. Project Plan	24
6.3. Budget	25
7. CONCLUSIONS	27
7.1. Revisiting the Objectives of the Investigation	27
7.2. Future Work	27
BIBLIOGRAPHY.	28

LIST OF FIGURES

3.1	Block Diagram of the Project Pipeline	8
3.2	Simple Training Example	14
3.3	Simple Prediction Example	14
4.1	MarginalES Price Time Series at 14:00H	17
4.2	MarginalES at 14:00H Data Distribution, Including Mean, Median and Mode Values	18
6.1	Gantt Diagram of the Project Lifecycle	24

LIST OF TABLES

6.1	Equipment Amortization	25
6.2	Human Costs	26
6.3	Final Cost Breakdown	26

1. INTRODUCTION TO THE PROBLEM

Energy plays an increasingly critical role in society, with reliable energy availability being essential for maintaining the modern high-functioning societies found across Europe, that benefit from a robust and advanced energy infrastructure. Furthermore, electricity is a strategic resource needed to succeed in the ecological transition that we are currently undergoing world wide, in which electric mobility is a key element of the process. All electric vehicles need convenient and easily available sources of energy to recharge. These can range from electric cars, to buses and trucks, all of which must be supported by accessible charging infrastructure to function effectively.

A important factor in this ever-growing necessity is the price of the available energy. But in recent times, that easy and affordable access has change. Across Europe, it has become more unpredictable ever since the commencement of the war between Russia and Ukraine, which marked and inflection point in the mostly cyclical nature of electricity prices. Prices have gone up considerably in the past three years, and it has been affecting everyone. From big consumers such as companies, to individual consumers, we all now pay a more expensive electricity bill at the end of the month. This has become a topic of great concern for consumers, as they more often reflect upon the current situation.

These concerns led me to think about pricing, exploring how it was structured, and its predictability from the perspective of an end consumer. Can we plan our use of energy according to the price? Can the price be predicted accurately? These were some of the questions I had that sparked my interest in looking into energy predictions.

After considering several modeling approaches, ranging from *Deep Learning* with long short-term memory (LSTM) neural networks, to statistical time-series specific models such as ARIMA, I chose to focus on traditional *Machine Learning* techniques. They offer a balance between predictive power and interpretability, building upon what was taught in the third year subject *Modern Theory of Detection and Estimation*. This decision was further supported by my tutor, Emilio Parrado Hernández, whose professional background in machine learning model design for investment banking shaped the direction of this work

The core of this project revolves around an innovative system design that integrates the use of a sliding window training approach for traditional machine learning models, such as Linear Regression, Lasso, and Random Forest, with the use of expressive features created from technical analysis indicators. This experimental use of technical analysis in the feature engineering phase is intended to investigate the effectiveness of these indicators applied in a different field. The idea is to aid the model training process by exposing it to different indicators that describe the trend, momentum and volatility of the time series.

2. BACKGROUND AND RELATED WORK

In Spain, the National Markets and Antitrust Commission, *Comisión Nacional de los Mercados y la Competencia (CNMC)* in Spanish, is the regulatory body in charge the functioning of the wholesale energy market, and the management and operations of the system. [1] Its methodologies must be followed by the designated energy market operator, the Iberian Energy Market Operator, *Operador del Mercado Ibérico de Energía (OMIE)* in Spanish.

This was decreed on October 1st 2004, in the Santiago de Compostela International Agreement between the Kingdom of Spain and the Portuguese Republic. In article 4, section 2, it is agreed upon that the "OMIP will act as the managing entity of the forward market and OMIE as the managing entity of the day market, following the previous compliance with the rules of the Party in which territory the head-office is located." [2]

The OMIE publishes a guide where they describe in depth the functioning of the daily energy market [3]. As the overseer of the market, they facilitate electricity transactions for the following day through offers from sellers and buyers. Every day at 12:00 CET, a market session determines electricity prices and energy volumes for the next day. Offers can be simple or complex, considering conditions like indivisibility, load gradient, minimum revenue, or scheduled shutdown.

The Iberian market has been coupled with the European market since 2014 for cohesion and integration purposes. Because of this, the OMIE's *Euphemia* algorithm is used to optimize economic surplus by matching supply and demand across different European markets, respecting market constraints, including interconnection capacity. After matching, results are sent to the System Operator, *Red Eléctrica de España (REE)* for validation to ensure technical feasibility. Adjustments may occur based on grid restrictions to ensure stability and efficient electricity distribution.

As a practical example, for a new producer to enter the daily energy market and sell electricity to the energy pool, they must take several key steps. The most essential one is registering as a producer for the Iberian electricity market, for which they will have to adhere to the current legal regulations and sign a contract to participate. They will then be able to submit offers to OMIE, which will result in increased competition. This increase in supply often leads to lower prices, specially if they introduce renewable energies, which have lower marginal costs compared to conventional energy sources such as nuclear or combined cycles.

2.1. Training a Non-Stationary Dataset with a Sliding Window Approach

Traditional machine learning models are usually trained on a fix set of data, that is split into various subsets to ensure proper training and evaluation. Starting with the *training set*, which is the largest portion of the dataset and is used to train the model. The algorithm learns patterns, weights, and relationships within the data during this phase. The model's internal parameters are adjusted based on this set.

The *validation set* is used during model development to fine-tune hyperparameters and assess the model's performance on unseen data. It helps detect overfitting, which is when a model performs well on the training data but poorly on new data. The validation set guides model optimization choices without touching the final evaluation set.

The *test set* is strictly separate from the others. It is held out until the end of the training process, to be used to evaluate the final model's generalization capability. It provides an unbiased estimate of how the model will perform on truly unseen data, simulating real-world deployment conditions.

In the context of this project, the dataset is an energy price time series, which is not directly compatible with the standard training methodology, as over time the dataset characteristics might shift, making an arbitrary training set useless. This is because the considered time series is non-stationary.

The proposed solution to mitigate the non-stationary nature of the dataset, is to progressively change our training set. The way of doing this is by creating a *sliding window* of data points that will compose the training set. This sliding window will determine the number of past data points that are relevant for the next prediction, since we will be effectively transforming the data set into a *stationary* one.

To work with this approach, the training set will be the entirety of the data points contained in the desired sliding window, and the test set will be the following row of data points (price values and features), which will be used to make the next prediction. Essentially, a new model will be created to make the following prediction, being discarded in favor of a new, slightly different model for the next prediction.

2.2. Machine Learning Models Selection

The model selection for the project will consist of three models: Linear Regression, Lasso Regression, and the Random Forest Regressor.

Linear Regression [4] is one of the most fundamental and widely used statistical techniques for modeling the relationship between a dependent variable and one or more independent variables. The method assumes a linear relationship between input features and the target, expressed as $y = X\beta + \varepsilon$, where β represents the model coefficients and ε is the error term. The model is trained by minimizing the residual sum of squares (RSS),

attempting to find the line that best fits the data. Due to its simplicity and interpretability, linear regression will serve as a baseline for our predictive modeling tasks, though it is limited in capturing non-linear relationships as it is mainly sensitive to multicollinearity.

Lasso Regression [5] (Least Absolute Shrinkage and Selection Operator) extends linear regression by introducing L1 regularization, meaning it penalizes the absolute magnitude of the regression coefficients. This penalty term, controlled by a hyperparameter α , encourages sparsity in the model by shrinking some coefficients up to zero, effectively performing feature selection. The optimization objective combines the ordinary least squares loss with the L1 penalty:

$$\hat{w} = \arg \min_w \left\{ \frac{1}{2n} \|y - Xw\|_2^2 + \alpha \|w\|_1 \right\}$$

This characteristic makes Lasso particularly useful in high-dimensional settings where many features may be irrelevant or redundant. While it can reduce overfitting and improve model generalization, it may also introduce bias by overly shrinking important coefficients, especially when predictors are highly correlated.

Finally, the *Random Forest Regressor* [6] is an ensemble learning method that constructs a collection of decision trees during training, and outputs the average prediction of the individual trees. Each tree in the "forest" is trained on a different sample of the data, and at each split, a random subset of features is considered, which promotes diversity among the trees. This approach helps mitigate the overfitting commonly associated with single decision trees, and improves predictive accuracy and robustness. Unlike linear or Lasso regression, Random Forest does not rely on assumptions of linearity or feature independence, making it well-suited for capturing complex, non-linear relationships in data. However, this flexibility comes at the cost of reduced interpretability compared to linear models.

2.3. Model Optimization: Hyperparameter Tuning and Feature Engineering

Hyperparameter tuning or *optimization* refers to the process of optimizing the model's training configuration parameters. These are parameters that are not learned from the data during the model training process, they are set before the training process begins and determine how the learning process of the model will develop. For *Lasso Regression*, α is a hyperparameter that controls the strength of the regularization. For the *Random Forest Regressor*, `n_estimators` (number of trees) and the `max_depth` (maximum depth of each tree), are a couple of its hyperparameters. These values for any given model, and the dataset employed in its training, must be experimented in order to find the values that result in the best model performance (e.g., lowest error, highest accuracy) [7].

TODO: Reference academic book: <https://www.microsoft.com/en-us/research/wp-content/uploads/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006>.

Feature Engineering is a crucial step in preparing, and expanding the dataset for training machine learning models. In this project, a particular approach involving *Feature Engineering with Technical Analysis* will be used, creating descriptive features based on the raw price data. Technical analysis, is a methodology for interpreting and analyzing price movements using statistical data and historical patterns to make more accurate predictions [8]. This involves generating technical indicators such as moving averages and indicators for momentum or volatility.

TODO: Reference academic book /article of Technical Analysis

These indicators are not just arbitrary transformations, they are intended to capture the inherent patterns and temporal information implicitly embedded inside the raw price series, theoretically reducing the need for complex time series models. The intention of generating these highly informative features, is to improve the robustness of the machine learning models, as the indicators should encode valuable predictive signals. This allows for a more straightforward model architecture while attempting to leverage the rich temporal dynamics of the data, as proven in other fields such as financial data analysis.

A variety of technical indicators such as *SMA*, *EMA*, *ROC*, *RSI*, *BBWidth* & *ATR* will be used. The *Simple Moving Average* will be used with a variety of window sizes. A shorter SMA could be useful to identify short-term trends in energy prices, capturing recent price movements, while a longer SMA could be useful to capture long-term trends in energy prices, helping identify the overall direction of the market. The *Exponential Moving Average* is similar to a simple moving average, but with the key difference being that it gives more weight to recent prices. This could be valuable to capture more immediate trends in energy prices. We will use the following *trend-following* indicators:

$SMA_3, SMA_5, SMA_7, SMA_{14}, SMA_{30}, SMA_{90}, SMA_{180}$

$EMA_3, EMA_5, EMA_7, EMA_{14}, EMA_{30}$

To measure momentum the price *Rate of Change* measures the percentage change between the current price and a price from a previous time period (e.g., 1 day, 7 days ago). It helps identify momentum in the market and is useful to highlight trends or sudden shifts in energy prices. Additionally, the *Relative Strength Index* is another momentum indicator. It is mainly a financial data point that can identify overbought or oversold conditions in the financial markets. Regarding its usage in the energy markets, the intention is it to track whether the price is approaching extreme values, which could indicate a reversal. We will use the following *momentum* indicators:

$ROC_3, ROC_5, ROC_7, ROC_{14}, ROC_{30}$

$RSI_5, RSI_7, RSI_{14}, RSI_{30}$

Finally, to measure volatility we employ indicators such as the Bollinger Band Width and the Standard Deviation. We will use the following *volatility* indicators:

$BBWidth_7, BBWidth_{14}$

$STD_7, STD_{14}, STD_{30}, STD_{90}$

2.4. Evaluation Metrics

To evaluate the predictive performance of the models, a combination of statistical metrics that quantify how closely the predictions align with actual values will be employed. Evaluating using multiple indicators ensures a robust and comprehensive assessment, mitigating the risk of overfitting to a single metric, and providing a deeper understanding of both average-case and worst-case predictive behavior. This multi-metric evaluation framework is essential for iteratively refining the model and selecting the optimal sliding window, and set of hyperparameters.

The *Mean Absolute Error* (MAE) serves as a primary accuracy metric throughout the model development. It calculates the average absolute difference between predicted and actual values, in the same units as the target variable (€/MWh in this context). It offers a simple interpretation of how much, on average, the predictions deviate from reality. Since the MAE treats positive and negative errors equally, it provides a balanced view of model accuracy, which is particularly useful for the hyperparameters tuning process across iterations of the same sliding window. Its simplicity makes it ideal for tracking the performance of each parameter as the and the model training advances along the time series. The formula for the MAE is defined as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where y_i is the true value, $\hat{y}_i$ is the predicted value, and n is the total number of observations.

The *Mean Squared Error* (MSE) serves as a second core performance indicator. Unlike the MAE, the MSE penalizes larger errors more heavily, by squaring the differences between predicted and actual values, highlighting outliers or poor predictions. This characteristic makes the MSE particularly effective in identifying models that may perform well on average, but fail under certain conditions. It is especially useful in diagnosing issues related to overfitting or inadequate generalization. The formula for the MSE is defined as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The *coefficient of determination*, denoted as R^2 , provides an intuitive measure of how well the model explains the variance in the target variable. An R^2 value close to 1 suggests that the model captures most of the variability in the data, while a value near 0 indicates a weak correlation between the input features and the target variable. This metric is particularly helpful when comparing multiple models or evaluating the effectiveness of

adding additional data points with feature engineering, as it gives a high-level summary of the overall model fit. The coefficient of determination (R^2) may be calculated with the following formula:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the observed values.

Beyond average errors, the prediction accuracy can be also evaluated in terms of *percentiles* to understand how well the model performs across different segments of the distribution. For instance, checking the 70th, 80th, and 90th percentiles of error allows for an assessment of both typical and extreme prediction behaviors. This adds granularity to the evaluation process, highlighting whether the model is consistently accurate or only performs well under specific conditions. The percentiles will be calculated as the top performers in a given range, meaning only the example, 70%, 80% or 90% of most accurate predictions would be considered.

Pivoting from the best percentiles, to the worst, the concept of *expectation shortfall*, commonly used in financial risk analysis, was included as a measure of tail risk in model predictions. This metric calculates the average prediction error among the worst-performing cases (e.g., the top 5% of errors), providing a quantitative estimate of the magnitude of extreme failures. In the context of energy prices, it helps us quantify how bad the prediction is when it fails, specially with over-estimations by hundreds of euros per MWh, since under-estimations shall be treated as 0, as we will work with the assumption that the price is not typically negative. This metric adds a risk-aware perspective to the model evaluation, ensuring that occasional large errors are accounted.

2.5. Related Work

Maybe comment about more typical case studies focusing on energy consumption, since that is something more consistent and predictable

As previously introduced, there are a variety of strategies to predict energy prices. There are various alternatives to machine learning, such as deep learning neural networks, or statistical time-series models. These are radically different approaches to the one taken in this project.

- Deep Learning for price predictions?
- LSTMs - Temporal Series?
- Energy consumption predictions?

3. PROPOSAL: SYSTEM DESCRIPTION

This chapter outlines the design of the system developed for the project, describing the complete *data processing pipeline* that was constructed. This will be explained starting from a blank state, detailing the complete set-up process required to run the project. Following this initial configuration, the data preparation stage will be outlined. This will be continued by the feature matrix creation process, concluding with the training of the machine learning models and the chosen evaluation method.

The following block diagram models the design of the system pipeline design in its most basic form:

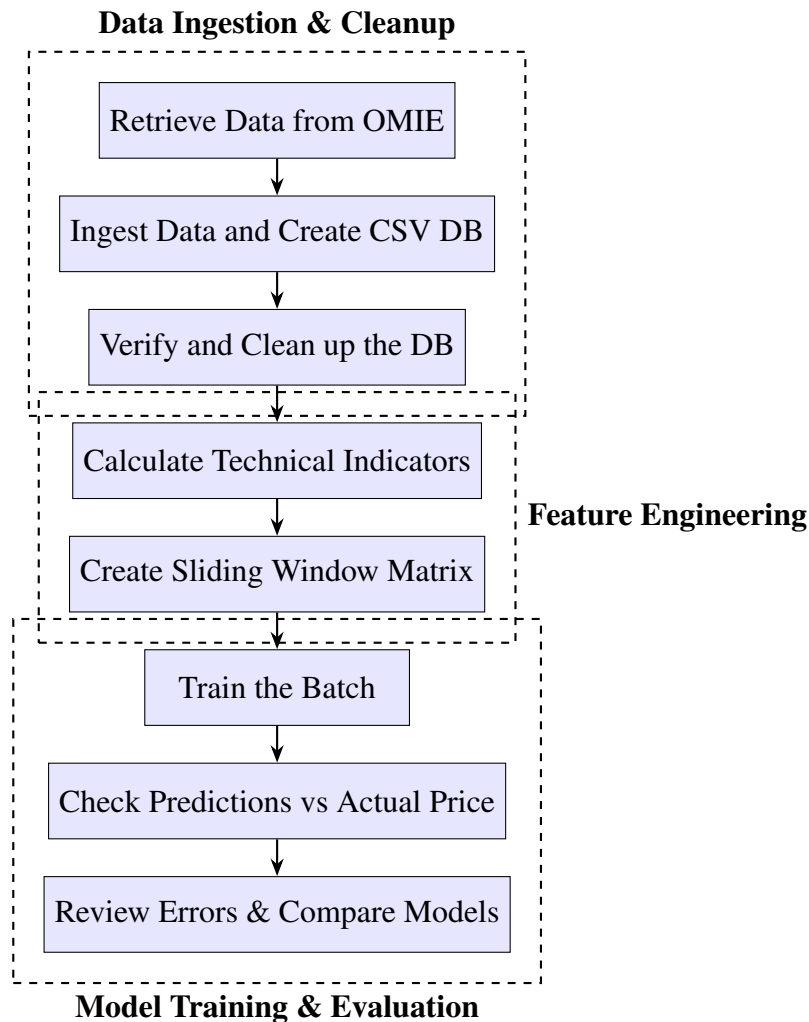


Fig. 3.1. Block Diagram of the Project Pipeline

3.1. Prior Set Up and Configuration

Before getting into the system explanation, there are some important pre-requisites that must be met in order to run the project. The project was developed and tested with Python 3.13 [9], so for any recreation of the project, it is highly recommendable to use the same version. For additional assurance of reproducibility, and as a universal best practice for Python development, it is strongly advisable to set up a new virtual environment [10]. Alternatively, another system that would achieve the same goal of reproducibility, and dependency conflict avoidance would be the use of Miniconda [11]. In this project, the built in Python module *venv* was chosen, with which a virtual environment can be easily created and configured with the project dependencies following the steps in *Appendix A*.

The chosen libraries were specifically selected to aid in the development of the project. Most of the project was written in Jupyter Notebooks, which are dependent on the *ipykernel* package. To aid in the retrieval of the data, the *requests* package was used to execute HTTP requests. For the manipulation of the data points, *pandas* was used to house the data structures, *numpy* for mathematical operations using arrays, and the *ta* technical analysis library was used to compute the technical indicators. Finally the machine learning algorithms were provided by the *scikit-learn* library.

Following the completion of this prior set up required for the project, the next sections will detail in depth the different steps involved in the system pipeline.

3.2. Data Preparation: Ingestion and Cleanup

The first step in any data preparation process would be to retrieve the raw data from the original source, in this case this means downloading it from the OMIE. It is publicly available data easily discoverable through their webpage [12]. The data collected was all that was initially available towards the end of 2024, with some extraordinary retrievals done later on in 2025 to further complete the set with more data points.

In this scenario, it meant retrieving data from 2018 onward. While for the 5 first years available the data was self contained in a compressed archive file, the rest from 2023 to our current 2025 was not. This meant that to retrieve the files, manually downloading each resource was required. In order to streamline this process, a short Python script was created to build the necessary URIs to retrieve the files with an HTTP GET request.

The Python script in *Appendix B* was written with modularity in mind for ease of use. It firstly creates the filenames according to the OMIE's file naming convention, inserting them into a list. This list is essential to successfully build the URIs with which the HTTP GET request will be made. Finally the function `download_files(urls)` makes the request, retrieving the files.

The newly downloaded raw data was manually organized into its corresponding folder per year. An example filename is the following: `marginalpdbc_20230102.1`. This is an important

organizational step for the ingestion of the data, as the next procedure involves a script that iterates through these folders, identifying all files that start with `marginalpdbc_`, and parsing them into an easier to use form, a Pandas DataFrame [13].

When a valid file is identified while iterating through the *year* folders, the script reads its content. It will perform some cleanup steps, such as removing the first and last rows which are a standard header and footer across all files. It then parses the remaining data, which is separated by semicolons, into a Pandas DataFrame for that specific day. It will name the columns to *Year*, *Month*, *Day*, *Hour*, *MarginalPT* (Portuguese Energy Price), and *MarginalES* (Spanish Energy Price). It will also filter out any rows where the *Year* column is not a number, ensuring data integrity. The columns are then converted to their appropriate data types, *integers* for date/time components and floats for marginal prices. Each processed daily dataset is then temporarily stored in a list.

After processing all the individual files, the script concatenates all the daily data from the list of DataFrames into a single, comprehensive DataFrame. It will then construct a *Datetime* column built from the *Year*, *Month*, *Day*, and *Hour* columns. It will also handle some peculiar cases where an *Hour* value of 25 might appear (likely due to daylight savings time zone adjustments) by converting it to 0. This *Datetime* column is then set as the DataFrame's index. Finally, it removes the individual date and hour columns as well as the *MarginalPT* column, leaving only the *MarginalES* as the primary target variable, along with the *Datetime* index.

This clean and combined dataset will be saved as a CSV file for ease of use, `raw_data.csv`. The script then performs further data cleanup by reloading the `raw_data.csv`, and sorting it by *Datetime*, saving it as `processed_data.csv`. The whole process concludes by verifying the completeness of the hourly timestamps within the processed data, generating the full range of hourly timestamps that should be there, and checking for any discrepancies. If no missing timestamps are found, it confirms successful data processing; otherwise, it reports the missing timestamps, indicating an error in the data collection or processing.

For the previous in-depth explanation, you may find in *Appendix C* the of Python script that was created for the desired data ingestion and cleanup tasks.

After completing this ingestion and cleanup process, the dataset will be reduced to just the 14th hour data point. The code in *Appendix D* will create a the final database file, which will house the singular *Datetime* value, and the *MarginalES* data point.

To aid in the predictions, the dataset is enhanced with more features. We can calculate a variety of different features to aid the training of the model. These can range from simple variables such as the day of the week, the month of the year, or if that day is a weekend or not, to more complex, such as technical indicators. By adding these auxiliary features derived from *technical analysis*, we can further expand the context that our models will be able to learn from, ideally improving their prediction accuracy capabilities.

To generate these additional features, and the strategically chosen technical indicators, the main database `processed_data.csv` is read into a DataFrame that will be progressively expanded with new feature columns. The resulting DataFrame will be saved to a secondary database `ta_metrics_hour_14.csv`. The code in *Appendix E* is an example of the process that was employed to calculate the different indicators, using the open source Python library *ta* [14].

3.3. Feature Matrix Creation

The creation of a *feature matrix* is a key step in the process of training any supervised, or unsupervised, machine learning model. In this context, the objective is to structure historical energy price data into a form that can be used effectively to predict future prices. This transformation of the time-series is critical to form our feature matrix, where observations must be ordered correctly to avoid losing the autocorrelation implicit in the series.

The intent of building the feature matrix consists on creating one unique matrix that can be used across any desired model. The feature matrix denoted as X , represents the input data used to train the model, while the target / label vector y contains the values the model is intended to learn how to predict.

The simple feature matrix of an example time series of 6 price values x_1 through to x_6 can be represented as:

$$X = \begin{Bmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{Bmatrix}, \quad y = \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \end{Bmatrix}$$

Where each row of X corresponds to a sliding window of past values, and its corresponding label y is the value that follows that window, which would be the actual energy price for the current day:

$$y_1 = x_4$$

$$y_2 = x_5$$

$$y_3 = x_6$$

This setup reflects the temporal dependency in the time-series data, where future values are predicted based on a certain number of past observations. This approach allows the machine learning models to "learn" these implicit patterns in the considered sliding window, enabling them to better predict the next value.

Incorporating technical analysis indicators into the feature matrix introduces additional information such as moving averages or the rate of change calculated over historical price windows. These enrich the model's inputs and are added as extra columns to the feature matrix:

$$X = \begin{Bmatrix} x_1 & x_2 & x_3 & ta_{11} & ta_{21} & \cdots & ta_{n1} \\ x_2 & x_3 & x_4 & ta_{12} & ta_{22} & \cdots & ta_{n2} \\ x_3 & x_4 & x_5 & ta_{13} & ta_{23} & \cdots & ta_{n3} \end{Bmatrix}$$

Where:

- x_i are raw price values from previous time steps,
- n is the number of indicators,
- ta_{ji} is the value of the j -th technical indicator computed based on the last price input of row i , i.e., x_{i+2} in this 3 previous days example.

The target / label vector will remain unchanged:

$$\mathbf{y} = \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \end{Bmatrix}$$

This design is motivated by the nature of the energy market data, which is *non-stationary*, as the statistical properties of electricity prices (e.g. mean, variance) change over time due to evolving supply and demand dynamics, regulatory interventions, and renewable energy contributions. Since the dataset has inherent *temporal dependencies*, as prices are typically dependent on recent values, making the assumption of independent and identically distributed (i.i.d.) samples would be invalid. The aim is to identify *local trends*, implementing a sliding windows approach allows the model to capture short-term patterns and trends without requiring the data to be stationary.

By using a sliding window mechanism, the models will be trained on a continuous stream of overlapping data segments. This not only increases the number of training samples, enhancing generalization, but also aligning with the real-world scenario of forecasting the next value based on a recent history of observations. The ultimate aim is to build feature matrix using the sliding window approach to provide a robust and flexible representation of temporal data, allowing traditional machine learning models to operate more effectively in a domain typically reserved for time-series-specific methods.

The Python Script found in *Appendix F* consists on the code necessary to create the simple feature matrix, which was initially generated based off *MarginalES* values, with the desired sliding window width for that training batch. For an example run we can retrieve from the database a sliding window of 6 days:

	Datetime	MarginalES
187	2019-01-01 14:00:00	65.88
188	2019-01-02 14:00:00	63.16
189	2019-01-03 14:00:00	66.70
190	2019-01-04 14:00:00	69.17
191	2019-01-05 14:00:00	64.00
192	2019-01-06 14:00:00	64.86

For which the resulting simple feature matrix is the following:

```
print(X_simple)
0      1      2
0  65.88  63.16  66.70
1  63.16  66.70  69.17
2  66.70  69.17  64.00

print(y_simple)
0    69.17
1    64.00
2    64.86
```

The simple feature matrix function was later modified and extended to incorporate additional feature columns such as the focus of our proposal, the technical indicators. These are the previously calculated features in order to expand the context our models will be able to train with. The script located in *Appendix G* demonstrates the flexible and auto-adaptable nature code, suitable for any number of inputted features (columns) obtained from the secondary feature enhanced database, which includes the additional technical indicators.

Furthermore, for an example run of the matrix creation with the technical indicators we can retrieve the extended database and select the same date range as in the previous example for comparison. Where we may observe an identical matrix to the earlier one, extended laterally to include the new feature columns, such as the technical indicators and additional "Datetime" related context:

```
print(X_extended)
```

	price_t-3	price_t-2	price_t-1	SMA_3	SMA_5	SMA_7	SMA_14	\
0	65.88	63.16	66.70	65.246667	64.974	63.842857	64.975000	
1	63.16	66.70	69.17	66.343333	66.026	64.490000	65.401429	
2	66.70	69.17	64.00	66.623333	65.782	65.434286	65.330000	

	SMA_30	SMA_90	SMA_180	...	RSI_30	BB_Width_7	BB_Width_14	\
0	64.886000	65.044333	67.227333	...	50.823486	17.871701	15.638302	
1	64.988333	65.038667	67.272889	...	52.153121	21.198596	16.528083	
2	64.947333	65.184222	67.267611	...	49.268675	11.634263	16.685668	

	STD_7	STD_14	STD_30	STD_90	month	day_of_week	is_weekend
0	3.080999	2.636139	2.620006	4.626188	1.0	3.0	0.0
1	3.691585	2.804414	2.726835	4.620755	1.0	4.0	0.0
2	2.055690	2.828060	2.732317	4.369895	1.0	5.0	1.0


```
print(y_extended)
```

0	69.17
1	64.00
2	64.86

3.4. Machine Learning Model Training

There are multiple strategies that can be implemented for an effective model training, depending on if the dataset is *stationary* or not. An *incremental* approach can be taken when the data is *stationary*, since in this case, the accumulating the data points will benefit the model's training. An *adaptive* approach is in order when the dataset is of *non-stationary* nature. Similar to an exponential moving average, where the newer data holds more weight, while the older data's importance is progressively minimized, this investigation's models were trained with a sliding window approach, where the older data points are cut off. This training method "forgets" the earliest values as they are no longer relevant in order to make the latest prediction, which is the case in the energy prices database

The idea behind the previously introduced *adaptive learning* approach is simple, we will create a 1 day step loop across the complete time series. A finite set of days will be used in training each model, with each instance being utilized for one single prediction. The training phase will consist on searching the optimal sliding window of days, with the calculated set of features. In each iteration, the model will be trained with a set feature matrix, and will then be fed with the next feature row to test its prediction abilities.

Continuing with our previous example, the model will be trained with the feature matrix X .

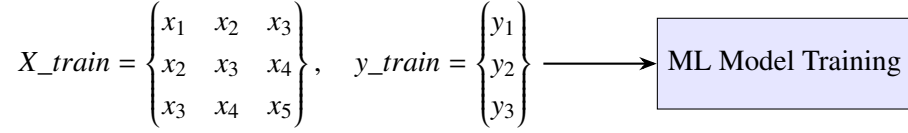


Fig. 3.2. Simple Training Example

In order to predict a new value, for example, $\hat{y}_4$, the model will be fed the next row of feature values $\{x_4, x_5, x_6\}$, where x_6 will be our last known value, y_3 .

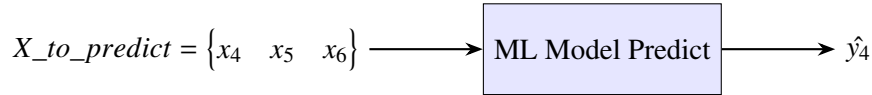


Fig. 3.3. Simple Prediction Example

The model selection for the project will consist of three models: the Linear Regressor, the Lasso Regressor, and the Random Forest Regressor. The first of the models, the *Linear Regressor* [4] will serve as our baseline. This model assumes that the target variable can be approximated as a linear combination of the input features. With the sliding window approach it will be possible to test multiple configurations of the model, but in comparison to the other tested models, it does not have an explicitly tunable hyperparameter.

Our second model, the *Lasso Regressor* [5], extends the linear model by introducing *L1 regularization*, meaning it has the ability to penalize unused features. This penalty encourages sparsity in the feature weights, effectively performing a feature selection. By tuning its hyperparameter *alpha* (α), we can modify the strength of the regularization. This can potentially improve the model's performance, by reducing overfitting, possibly improving its generalization capabilities. To optimize its predictive accuracy, we will later experiment across a variety of *alpha* hyperparameter values.

The third and final model considered in this investigation, is the *Random Forest Regressor* [6], an ensemble learning method based on decision trees. During training, it constructs multiple decision trees, and aggregates their outputs by averaging the regression tasks. We can tune its training process by adjusting parameters such as *n_estimators* (number of trees) and the *max_depth* (maximum depth of each tree). Its output will be averaged, which can be a limiting factor for its predictions, since even if it is capable to learn more complex patterns than the aforementioned models, it will tend to underestimate extreme peak values. This model has been included as a non-linear alternative to validate the performance of the linear models and capture more complex relationships within the data.

Regardless of the chosen model, the training strategy will be as follows: After importing the complete dataset, we will define the number of days you want to include in the sliding window. This determines how many days from the time series will be included in the training. At the same point, we will select the desired number of previous days to be included in the feature matrix as columns. Depending on the selection of the sliding window size, there will be more or less models to train.

After initializing the variables where the predictions, actual values, and their time stamps will be stored at, the training loop will begin. Here, a new model will be trained per iteration as seen in Figure 3.2, and a new value will be predicted as seen in Figure 3.3. Once all models and predictions have been processed, the `prediction_df` DataFrame will be available for evaluation of the predictive precision of the training loop.

To finalize the system description, the example script found in *Appendix H* shows in Python code the basic concept of our sliding window training process, in which any given model could be used. However, it is a basic example that does not test multiple sliding windows, nor consider any hyperparameter tuning. This will be covered in the following section, in the dedicated *Experimentation* chapter.

4. EXPERIMENTATION

This chapter is dedicated to presenting all of the relevant information pertaining the experimentation phase of the project. We review the specifics of the data that was dealt with, and explain the decisions taken for the preparation of such data, and the training of the different machine learning models employed. We review all of the results of the different models, and discuss in depth the different set-up variations for the models.

4.1. Data Description

The data that was used for this project was exclusively publicly available information, published by the OMIE (*Operador del Mercado Ibérico de Energía*), the operator of the Iberian energy market. The data retrieved from their website [12] for this project takes the following shape:

```
MARGINALPDBC;  
2018;01;01;1;28.1;6.74;  
2018;01;01;2;33;4.74;  
2018;01;01;3;32.9;3.66;  
...  
2018;01;01;22;28.1;21.95;  
2018;01;01;23;28.1;23.52;  
2018;01;01;24;27.6;16.35;  
*
```

We can understand and interpret this format following the OMIE's guide: *Modelo de Ficheros para la distribución pública de Información del mercado de electricidad 1.35* [15]. In page number 67, chapter 6.18 we may find the following information regarding the encoding format for the information:

6.18 Precios marginales del mercado diario (MARGINALPDBC)

Fichero con los precios marginales del Mercado Diario para cada una de las horas.

Nombre del fichero: `marginalpdbc_aaaammdd.v` donde `aaaammdd` corresponde a la fecha de sesión y `v` es la versión del fichero.

Descripción de los campos:

CAMPO DESCRIPCIÓN VALORES VÁLIDOS

Año Año I4 - 20XX

Mes Mes I2 - 1 a 12

Día Día I2 - 1 a 31

Hora Hora I2 - 1 a 25

MarginalPT Precio marginal zona Portuguesa F8.2 - -99999.99 a 99999.99
Marginales Precio marginal zona Española F8.2 - -99999.99 a 99999.99

From this data description we can obtain the following column names: *Año*, *Mes*, *Día*, *Hora*, *MarginalPT* and *MarginalES*. As a proof of concept, in this investigation we will be focusing on a single time slot, simplifying the data, from multiple data points per day, to a single one. We are exclusively interested in the value of the last column, *MarginalES*, and specifically the row pertaining the 14th hour of each day.

Dataset Analysis and Interpretation

After the data is ingested to the database, we can inspect the time series by visualizing the data points, and the data distribution, including the mean, median, and mode values of the dataset.

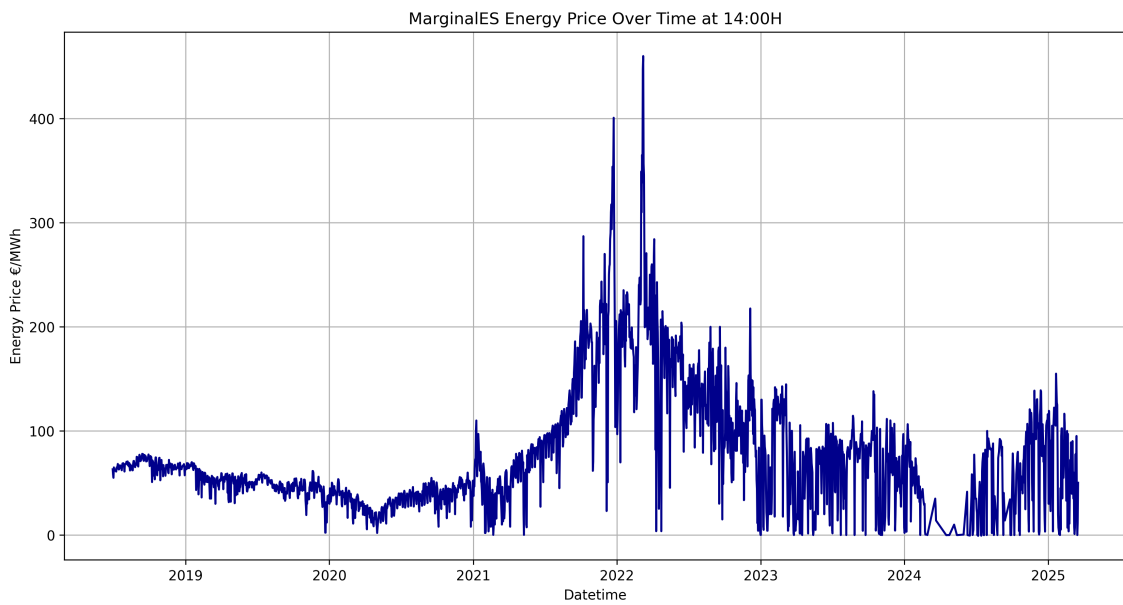


Fig. 4.1. MarginalES Price Time Series at 14:00H

Visual inspection of the dataset reveals that the time series does not meet the assumptions of being independently and identically distributed (i.i.d.) and is also clearly non-stationary. The series exhibits temporal dependencies, with strong correlations between consecutive observations, thereby violating the independence assumption essential for many statistical and machine learning models.

The energy prices show distinct periods, starting with relatively stable prices around 50-70 €/MWh from 2018 to late 2020, followed by growing volatility and a dramatic surge beginning in late 2021 that peaks at 460 €/MWh in 2022. This price escalation coincides with the escalating tensions, and eventual start of the war between Russia and Ukraine in February 2022, which severely disrupted European natural gas supplies and created unprecedented energy market volatility.

Since natural gas is a crucial input for electricity generation, this geopolitical crisis directly impacted electricity prices by disrupting the matching of supply and demand in OMIE's daily

market, where gas-fired power plants often set the marginal price. The subsequent gradual decline in 2023-2024 reflects market adaptation with alternative supply arrangements, and a reduced gas dependency, though prices remain elevated compared to pre-war levels.

These clear shifts in behavior over time show that the data is not stable, as the average values, variability, and overall distribution change across different periods. Because of this, it makes sense to use an *adaptive* sliding window approach, where the model is regularly retrained on the most recent data. This helps it stay up to date with the latest trends and better handle the ups and downs of a changing energy market.

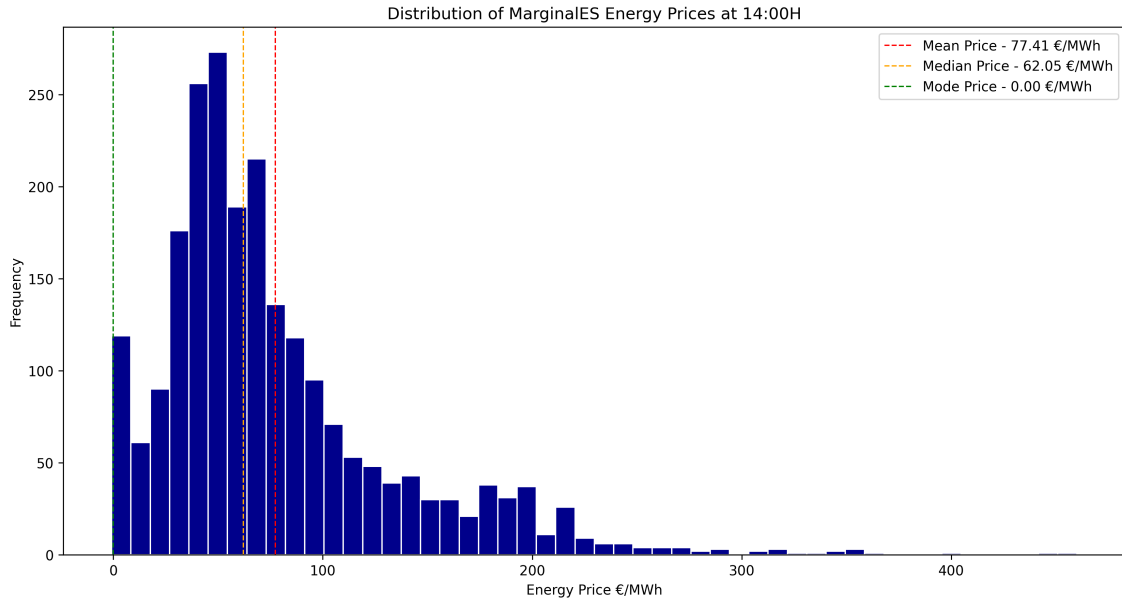


Fig. 4.2. MarginalES at 14:00H Data Distribution, Including Mean, Median and Mode Values

An interesting observation to comment upon is the mode value, which as seen in the graph, is zero. This can be attributed to the selected hour of the day for our dataset being 14:00H, which is usually around the time when the sun will be at its highest point in the sky, marking the peak production time for solar photovoltaic energy in Spain. And since Spain has an extremely large amount of solar power plants, this means that there will be an overabundance of energy, and therefore the price will plummet down to 0.

In general, the time series possesses a high correlation between the prices of neighboring days, with seasonal variations, slow and progressively changing the price. The task at hand for the models is to figure out the best fit to the time series for any of the proposed sliding window configurations.

4.2. Model Set Up Explanation

How do I use in practice the previously explained system - what parameters did I modify to obtain the final results. Feature engineering, Alpha variation testing in Lasso, Tree depth and number of leaves.

As a ground truth

Que modelos y que configs

Cross validation?? Not really done, in another way w the sliding window

cambio parametros pero no la arquitectura del modelo

Linear Regression

A linear combination of every feature

No optimization to be done here, just testing different sliding windows

Lasso Regression

A linear model that penalizes unused features

Experimenting with alpha using a loop or GridSearchCV to find the optimal value.

As mentioned earlier, tuning the hyperparameter *alpha* enables To optimize our predictions, we will later experiment across a variety of *alpha* hyperparameter values.

To achieve this, all of the parameter values selected will be tested in every sliding window iteration, noting down the *alpha* value that renders the result with the minimum error to the actual data point.

Random Forest

Soft on peaks - tune hyperparameters like n estimators, max depth, etc.

There are two main parameters that we will focus on tuning. The number of trees in the forest (*n_estimators*) and the maximum depth of the tree branches (*max_depth*) will be tuned.

Similar to the training process of the Lasso Regressor, for every sliding window iteration each combination of parameters will be tested. Likewise, we will save the combination of parameters with the best result, with the least error to the actual data point.

4.3. Results of the Testing

- Show best set of example results in graphs for each model
- Create one unique table comparing the best iteration of each technology
- Following the table, a simple and easy comparison commenting on the best sliding window, and the impact of tuning the model hyperparameters

4.4. Discussion

- This would be the final explanation that I would give to a colleague of mine, with full details on how everything performed, and how the batch runs stack up.
- Create one table per model to compare across the other not displayed iterations.
- Deep dive into the error rates of each run and compare with other iterations/ batches and

other models:

MAE, MSE, R2, Percentiles & Expectation Shortfall

5. REGULATORY FRAMEWORK

In any data-driven project understanding the regulatory landscape is essential. This chapter addresses the legal, ethical, and operational constraints that governed the use of data and software throughout the development of this project. From the data acquisition and pre-processing, to the software deployment and reproducibility, each component aims to comply with relevant national and international standards to ensure a responsible and legal development.

5.1. Data Availability

The dataset used in this project was obtained from the OMIE, a publicly traded company designated as the Iberian energy market operator. This organization publishes the energy market data as part of their legal obligations set by the Spanish CNMC for transparency, aligned with European regulations.

For the purpose of this thesis, which is strictly academic and investigative in nature, Spanish and European regulations allow for the use of publicly available institutional data without requiring additional licenses. This includes data shared under open access or public sector information frameworks, specifically for research, education, or non-commercial work.

However, it is important to note that if the project were to evolve into a commercial application, the legal requirements would change. In such a scenario, additional licensing agreements, permissions for commercial use, and compliance with intellectual property laws may be necessary.

While no personal or sensitive data is used in this project, future developments that incorporate customer or proprietary data would also need to comply with GDPR and ethical standards for responsible AI and data use, especially in markets as sensitive as energy trading.

5.2. Software & Licenses

This project was developed using a combination of open-source and proprietary software tools. In the context of regulatory and legal frameworks, the selection and usage of these tools are not only technical decisions, they also bear implications related to licensing compliance.

To ensure transparency and reproducibility, all core dependencies have been documented. Where applicable, licensing schemes have been reviewed to ensure compatibility with both academic and industrial research standards.

For clarity, the software tools have been categorized into two distinct parts: required software to reproduce the project, and optional tools to enhance the development workflow.

Required Software:

- Python: Primary programming language used throughout the project. Python is open source under the Python Software Foundation License [16].

- Pandas: Data manipulation library, distributed under a variety of permissive open-source licenses [17].
- NumPy: Numerical computing library, distributed under its own open-source license [18].
- Scikit-learn: Machine learning library, used for model training and evaluation. [19].
- ta (Technical Analysis) Library: Used for calculating financial indicators to aid in the time series analysis. It is available under the MIT license, a permissive open-source license [14].
- Matplotlib: Used for data visualization, distributed under a permissive open source license [20].
- macOS: Closed-source operating system on which development was performed. Its usage is subject to commercial a licensing agreement by Apple Inc. [21].

Optional Software:

- Git: Open-source version control system licensed under GPLv2, used for code tracking and collaboration [22].
- Git: Open-source distributed version control system, used for code tracking and collaboration [22].
- GitHub: Repository hosting platform for Git repositories. Though it hosts open-source projects, GitHub itself is a proprietary service governed by its own Terms of Service [23]
- Visual Studio Code: Open-source code editor under the MIT license, provided by Microsoft [24].
- Visual Studio Code Extension: Microsoft Data Wrangler: A proprietary extension that provides pandas.DataFrame visualization capabilities during development [25].
- Python Virtual Environment (venv): Used to make the project environment reproducible. It is best practice to create new Python virtual environments for each new project, specially in a production environment where certain versions are required in order for all components to work as intended [10]. An alternative method of dealing with this would be using Conda, and creating with it a new Conda environment [11].
- Overleaf: Cloud-based LaTeX editor used to compile this document. Overleaf is a closed-source platform with cloud-hosting considerations that may have implications under data protection regulations such as the GDPR [26]
- LaTeX: Open-source typesetting language used to produce the final project document. Licensed under its own LaTeX Project Public License [27].

No sé como comentar el uso de IA:

OpenAI ChatGPT, Google Gemini & Anthropic Claude: closed source large language models used to aid in troubleshooting general coding issues

Relleno o paso de esto: <https://uc3m.libguides.com/en/TFG/GenAI> ?

No veo formato para Latex, sólo un word. Igualmente para rellenarlo con: si, la he usado, pero sólo para lo que se permite en el reglamento (la pagina web)... Pues no sé si merece la pena, o cuan de estrictos serán.

En la matriz de evaluacion no figura como obligatoria: https://www.uc3m.es/secretaria-virtual/media/secretaria-virtual/doc/archivo/doc_matriz_evaluacion/eps_matrizevaluacion_tfg_definitiva_nov24__espanol_ingles.pdf

Simplemente aquí dice que la incluya si la uso: https://www.uc3m.es/ss/Satellite/SecretariaVirtual/es/TextoMixta/1371210936260/Trabajo_de_Fin_de_Grado#defensa

6. SOCIO-ECONOMIC ENVIRONMENT

Energy markets play a central role in modern economies, directly affecting industrial output, consumer costs, and overall economic stability. Electricity is a resource that cannot be easily stored at scale. Its demand fluctuates continuously, and supply must adapt almost instantaneously. As a result, accurate energy price prediction is not just a technical challenge, but a key enabler for operational efficiency, market stability, and policy decision-making.

This project operates within this socio-economic context. Understanding the broader environment highlights its relevance and the potential future applications of the work. The forecasting models developed here could contribute to better-informed decision making across multiple stakeholders, from grid operators and energy producers to regulators, businesses, and end consumers.

6.1. Impact

Why is this relevant?

- Energy Prediction
- Shutdowns
- Price sensitivity

In this project we have

6.2. Project Plan

TODO: Gantt diagram representing time spent in each phase - Design and innovation time frames

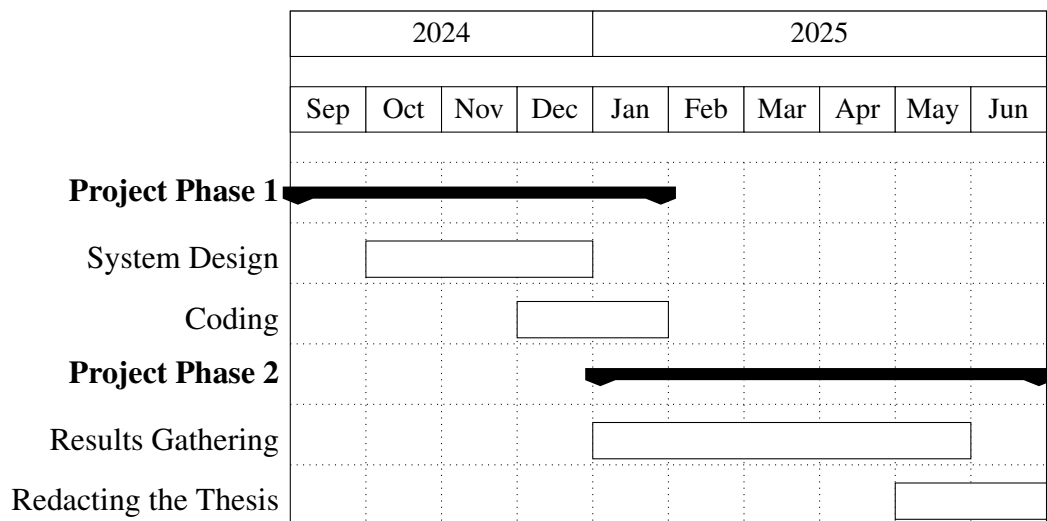


Fig. 6.1. Gantt Diagram of the Project Lifecycle

6.3. Budget

This section will consist of a complete price breakdown of the investigation, consisting of material costs such as the equipment utilized, and the human capital that composed the investigation team. It is notable to highlight that as previously mentioned, no paid license of any sort was obtained for the realization of this project. All of the programs, tools or code, were either *Open Source* or free to use software under exclusive *non-commercial use* licenses.

The most basic requirement for the physical needs of this project is a computer with an desktop operating system. Since the project was written in Python [16], it would be reasonable to consider it platform agnostic. Any computer capable of running any recent operating system, configured with a recent version of Python would be able to run this project. This includes any of the three most popular desktop operating systems, Windows, macOS and any Linux distribution.

The project was mainly tested using Python 3.13 [9] which does require a recent operating system such as macOS 13 or Windows 10. But as for the project itself, there is no platform, processor architecture, or processing power requirements. In my case, I mainly developed the code in my personal laptop, a 2021 MacBook Pro configured with the ARM M1 Pro chip and 16GB of RAM, running the latest available macOS version at this time, macOS 15 [21]. This laptop was obtained from a second-hand store circa June 2024, mainly for personal use, but also helping keep the theoretical project costs down.

The following table outlines the specific equipment costs incurred for the development of this investigative project:

Equipment	Price (€)	Amortization per year (€)	Useful Life (years)	Usage Time (months)	Cost (€)
MacBook Pro M1 Pro	1250	312.5	4	9.5	247.4

TABLE 6.1. EQUIPMENT AMORTIZATION

Regarding the human capital that was required for the successful development of this project, it will consist exclusively on two people. The director of my bachelor's thesis, professor Emilio Parrado Hernández, PhD, as the expert Machine Learning consultant, and myself as a junior software engineer.

The following table displays the estimated hourly rates of each person, the hours dedicated to the project, and the total cost:

Name	Role	Price (€/hour)	Hours	Cost (€)
Emilio Parrado Hernández	Expert ML Consultant (PhD)	200	-	-
Rodrigo De Lama Fernández	Junior Engineer (Pre-Graduate)	30	-	-

TABLE 6.2. HUMAN COSTS

For our final budgeting considerations, other miscellaneous expenses, such as transportation costs to the university for on site meetings with my bachelor's thesis director. Summing up all costs, the following table estimates what this project would have cost to investigate:

Concept	Cost (€)
Direct/Material Costs	247.4
Engineering Costs	-
University Carlos III Costs - Expert ML Consultant	-
Total	-

TABLE 6.3. FINAL COST BREAKDOWN

7. CONCLUSIONS

In this investigation we have analyzed in depth an innovative methodology for predictive modeling of energy prices with Machine Learning algorithms, enhanced by the usage of technical analysis indicators in feature engineering, in order to enhance prediction precision. This final chapter will review the general conclusions attained throughout the development of this project's Machine Learning based solution for energy price predictions. It will also discuss potential development paths for future work pertaining the system designed in this investigative project

7.1. Revisiting the Objectives of the Investigation

After finishing the whole project, read it, and reintroduce the objectives to the reader. Remind them of the completion of them

Predicting the prices in an accurate manner using Machine Learning models using feature engineering with technical analysis indicators

7.2. Future Work

There are several promising directions to improve and extend this investigative work. Firstly, future efforts should focus on reducing the overall absolute error of the model, while also addressing the isolated error spikes that occasionally lead to significant deviations in prediction accuracy. Achieving smoother and more stable forecasts would result in better percentile accuracy across the entire distribution of predicted prices.

Another potential avenue is the development of a hybrid model architecture that combines the strengths of different algorithms. For example, integrating the consistency and robustness of Random Forests with the sensitivity of linear models would potentially enable capturing sharp price variations and peaks, creating a more balanced and accurate predictor.

Finally, with further validation and performance tuning, this project could evolve into a production-ready system, potentially serving as the foundation for a commercial application capable of providing real-time electricity price forecasts to companies, traders, and consumers seeking to optimize their energy costs.

BIBLIOGRAPHY

- [1] Jefatura del Estado, *Circular 3/2019, de 20 de noviembre, de la comisión nacional de los mercados y la competencia, por la que se establecen las metodologías que regulan el funcionamiento del mercado mayorista de electricidad y la gestión de la operación del sistema*. https://www.boe.es/diario_boe/txt.php?id=BOE-A-2019-17287, Dec. 2019.
- [2] Government of Portugal and Government of Spain, *International agreement between the portuguese republic and the kingdom of spain for the creation of an iberian electricity market (mibel)*, https://www.omip.pt/system/files/2020-01/santiago_international_agreement_01.oct.2004_eng_01_0.pdf, Specifically, Article 4: Creation of an Iberian Market Operator, Oct. 2004.
- [3] OMIE, *Funcionamiento del mercado diario*, https://www.omie.es/sites/default/files/inline-files/mercado_diario.pdf.
- [4] The scikit-learn Development Team, *Scikit-learn: Linear regression*, https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html.
- [5] The scikit-learn Development Team, *Scikit-learn: Lasso regression*, https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Lasso.html.
- [6] The scikit-learn Development Team, *Scikit-learn: Random forest regressor*, <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>.
- [7] Amazon Web Services, *What is hyperparameter tuning?* <https://aws.amazon.com/what-is/hyperparameter-tuning/>.
- [8] K. Montevirgen. “Technical analysis: Is there predictive power in chart data?” (Sep. 16, 2022), [Online]. Available: <https://www.britannica.com/money/what-is-technical-analysis>.
- [9] Python Software Foundation, *Python 3.13*, <https://www.python.org/downloads/release/python-3130/>, 2025.
- [10] Python Software Foundation, *Venv — creation of virtual environments*, <https://docs.python.org/3/library/venv.html>, 2025.
- [11] Anaconda Inc., *Miniconda*, <https://www.anaconda.com/docs/getting-started/miniconda/main>, 2025.

- [12] OMIE, *Precios horarios del mercado diario en españa*, <https://www.omie.es/es/file-access-list?parents=/Mercado%20Diario/1.%20Precios&dir=Precios%20horarios%20del%20mercado%20diario%20en%20Espa%F1a&readdir=marginalpdbc>, Accessed on June 11, 2025, Jun. 2025.
- [13] The Pandas Development Team, *Pandas dataframe*, <https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.html>, 2025.
- [14] bukosabino, *Technical analysis library in python (ta)*, <https://github.com/bukosabino/ta>, 2025.
- [15] OMIE, *Modelo de ficheros para la distribución pública de información del mercado de electricidad 1.35*, https://www.omie.es/sites/default/files/2024-06/Formato_ficheros_inf_pub_135_0.pdf, Accessed on June 11, 2025, Jun. 2024.
- [16] Python Software Foundation, *Python programming language*, <https://www.python.org/about/>, 2025.
- [17] The Pandas Development Team, *Pandas: Python data analysis library*, <https://pandas.pydata.org>, 2025.
- [18] The NumPy Development Team, *Numpy: Array programming for scientific computing*, <https://numpy.org>, 2025.
- [19] The scikit-learn Development Team, *Scikit-learn: Machine learning in python*, <https://scikit-learn.org/stable/>, 2025.
- [20] Matplotlib Development Team, *Matplotlib: A comprehensive library for creating static, animated, and interactive visualizations in Python*, <https://matplotlib.org/>.
- [21] Apple Inc., *Macos sequoia*, <https://www.apple.com/macOS/macOS-sequoia/>, 2025.
- [22] Git, *Git: A distributed version control system*, <https://git-scm.com>, 2025.
- [23] GitHub, *Github: Repository hosting platform*, <https://github.com/about>, 2025.
- [24] Microsoft Corporation, *Visual studio code*, <https://code.visualstudio.com/docs/editor/whyvscode>, Open source code editor, 2025.
- [25] Microsoft, *Data Wrangler: A code-centric data viewing and cleaning tool for Visual Studio Code*, <https://marketplace.visualstudio.com/items?itemName=ms-toolsai.datawrangler>.
- [26] Overleaf, *Overleaf: Online L^AT_EX editor*, <https://www.overleaf.com/about>, 2025.
- [27] LaTeX Project, *Latex: A document preparation system*, <https://www.latex-project.org>, 2025.

APPENDIX A: PROJECT ENVIRONMENT CONFIGURATION

Create the virtual environment:

```
python -m venv .venv
```

Activating the new virtual environment:

in macOS

```
source .venv/bin/activate
```

in Windows

```
.venv\Scripts\Activate.ps1
```

To reproduce the exact environment used in the project, a *requirements.txt* file must be created with the following contents:

```
ipykernel==6.29.5  
matplotlib==3.10.1  
numpy==2.2.3  
pandas==2.2.3  
requests==2.32.3  
scikit-learn==1.6.1  
ta==0.11.0
```

To install the dependencies run:

```
pip install -r requirements.txt
```


APPENDIX B: AUTOMATED DOWNLOAD OF OMIE RAW DATA

```
1 import requests
2 from datetime import datetime, timedelta
3
4 # Usage instructions:
5     # Set file name to save the filenames to download
6     # Set start and end dates
7
8 file_path = 'to_download.txt'
9 start_date = datetime(2023, 1, 1)
10 end_date = datetime(2025, 3, 18)
11
12 # Compose filenames to download
13 def compose_filenames():
14     start_date = start_date
15     end_date = end_date
16
17     # Generate the entries for the number of days comprehended
18     to_generate = (end_date - start_date).days + 1
19     entries = []
20     for i in range(0, to_generate): # x days from start to end
21         date_entry = start_date + timedelta(days=i)
22         entries.append(f'marginalpdbc_{date_entry.strftime("%Y%m%d")}
23                        }.1')
24
25     # Save to a .txt file
26     with open(file_path, 'w') as file:
27         for entry in entries:
28             file.write(entry + '\n')
29
30 # Example URI to build
31 # https://www.omie.es/es/file-download?parents%5B0%5D=marginalpdbc&
32 # filename=marginalpdbc_20230102.1
33 def read_filenames_and_compose_urls(file_path):
34     base_url = "https://www.omie.es/es/file-download?parents%5B0%5D=
35     marginalpdbc&filename="
36
37     # Read the filenames from the .txt file
38     with open(file_path, 'r') as file:
39         filenames = [line.strip() for line in file if line.strip()]
40
41     # Compose the URLs
42     urls = [base_url + filename for filename in filenames]
```

```
43 def download_files(urls):
44     for url in urls:
45         # Extract the filename from the URL
46         filename = url.split('=')[-1]
47
48         # Send a GET request to download the file
49         response = requests.get(url)
50
51         # Check if the request was successful and write to file
52         if response.status_code == 200:
53             # Write the content to a file with the extracted
54             # filename
55             with open(filename, 'wb') as file:
56                 file.write(response.content)
57             print(f"Downloaded {filename}")
58         else:
59             print(f"Failed to download {filename}")
60
61 # Execution
62 compose_filenames()
63 urls = read_filenames_and_compose_urls(file_path)
64 download_files(urls)
```

APPENDIX C: DATA INGESTION AND CLEANUP

```
1 import pandas as pd
2 import os
3
4 # Base path to the folder containing the year by year data folders
5 base_path = '../TFG/data/'
6
7 # Initialize an empty list to store DataFrames
8 all_data = []
9
10 # Iterate through each year folder and process all files
11 for root, dirs, files in os.walk(base_path):
12     for file in files:
13         # Check if the file starts with 'marginalpdbc_' (ignoring
14         # the rest, including the extension)
15         if file.startswith('marginalpdbc_'):
16
17             # Read the file
18             file_path = os.path.join(root, file)
19             with open(file_path, 'r', encoding='utf-8') as f:
20                 lines = f.readlines()
21
22             # Remove the first line (MARGINALPDBC;) and the last
23             # line (*)
24             data_lines = lines[1:-1]
25
26             # Convert the list of semicolon-separated strings into a
27             # DataFrame
28             daily_data = pd.DataFrame([line.strip().split(';') for
29                                     line in data_lines])
30
31             # Remove trailing ';' from the last column
32             if daily_data.shape[1] > 6:
33                 # Drop empty columns
34                 daily_data = daily_data.iloc[:, :6]
35
36             # Assign column names
37             daily_data.columns = ['Year', 'Month', 'Day', 'Hour', '
38                                 MarginalPT', 'MarginalES']
39
40             # Remove rows with invalid data (in case accidental
41             # empty rows, or non-numeric values, get included)
42             daily_data = daily_data[daily_data['Year'].str.isdigit()
43                                     ] # Only keep rows where 'Year' is a number
44
45             # Convert the columns to the appropriate data types
```

```

39         daily_data = daily_data.astype({
40             'Year': int, 'Month': int, 'Day': int, 'Hour': int,
41             'MarginalPT': float, 'MarginalES': float
42         })
43
44         # Append daily data to the list
45         all_data.append(daily_data)
46
47     # Concatenate all daily DataFrames into one big DataFrame
48     full_data = pd.concat(all_data, ignore_index=True)
49
50     # Create a datetime column from Year, Month, Day, and Hour
51     # Adjust the 'Hour' column to deal with the potential 25th hour
52     # issue in case of timezone adjustment
53     full_data['Hour'] = full_data['Hour'].apply(lambda x: 0 if x == 25
54         else x)
55     full_data['Datetime'] = pd.to_datetime(full_data[['Year', 'Month', '
56         Day', 'Hour']], errors='coerce')
57
58     # Set the 'Datetime' column as the index
59     full_data.set_index('Datetime', inplace=True)
60
61     # Drop the now unnecessary time columns and 'MarginalPT', as we are
62     # focusing on the Spanish market
63     full_data.drop(['Year', 'Month', 'Day', 'Hour', 'MarginalPT'], axis
64         =1, inplace=True)
65
66     # Save the full dataset to a CSV file
67     full_data.to_csv('../data/raw_data.csv')
68
69     # Post-ingest database cleanup:
70     # Load and sort the data by the 'Datetime', and save the sorted data
71     # to a new file
72     df = pd.read_csv('../data/raw_data.csv')
73     df = df.sort_values(by='Datetime')
74     df.to_csv('../data/processed_data.csv', index=False)
75
76     # Read the CSV file
77     df = pd.read_csv('../data/processed_data.csv')
78     df['Datetime'] = pd.to_datetime(df['Datetime']) # Parse 'Datetime'
79     # column to datetime objects
80
81     # Generate the complete range of hourly timestamps
82     full_range = pd.date_range(start=df['Datetime'].min(), end=df['
83         Datetime'].max(), freq='h')
84
85     # Find any missing timestamp
86     missing_timestamps = full_range.difference(df['Datetime'])
87
88     # If successful, print a message
89     if missing_timestamps.empty:

```

```
81     print("Data ingestion completed successfully.")
82
83 # Else, print an error message
84 else:
85     print("Error: Missing timestamps found. Check the data.")
86     print("Missing timestamps:\n missing_timestamps")
87
88     print("Data ingestion completed with errors.")
```

APPENDIX D: DATASET REDUCTION TO THE 14TH HOUR

```
1 import pandas as pd
2
3 # Load the original dataset
4 data_path = '../..data/processed_data.csv'
5 df = pd.read_csv(data_path, parse_dates=['Datetime'])
6
7 # Define the hour to predict and filter down to the 14:00H data
  point
8 hour_to_predict = 14
9 df_hour = df[df['Datetime'].dt.hour == hour_to_predict].copy()
10
11 # Save the final database
12 df_hour.to_csv("../..data/hour_14_metrics.csv", index=False)
```

APPENDIX E: TECHNICAL ANALYSIS INDICATORS CALCULATIONS

```
1 import pandas as pd
2 import ta # bukosabino's Technical Analysis Library
3 import numpy as np
4
5 # Load the data
6 data_path = 'data/processed_data.csv'
7 df = pd.read_csv(data_path, parse_dates=['Datetime'])
8
9 # Reduce dataset to only to the 14th hour
10 hour_to_predict = 14
11 df = df[df['Datetime'].dt.hour == hour_to_predict].copy()
12
13 # The "window" parameter is measured in days
14
15 # Trend-following - SMA and EMA
16 # Simple Moving Average
17 df[f'SMA_{3}'] = ta.trend.SMAIndicator(close=df['MarginalES'],
18 window=3).sma_indicator()
19 df[f'SMA_{5}'] = ta.trend.SMAIndicator(close=df['MarginalES'],
20 window=5).sma_indicator()
21 df[f'SMA_{7}'] = ta.trend.SMAIndicator(close=df['MarginalES'],
22 window=7).sma_indicator() # past week trend
23 df[f'SMA_{14}'] = ta.trend.SMAIndicator(close=df['MarginalES'],
24 window=14).sma_indicator()
25 df[f'SMA_{30}'] = ta.trend.SMAIndicator(close=df['MarginalES'],
26 window=30).sma_indicator() # past month trend
27 df[f'SMA_{90}'] = ta.trend.SMAIndicator(close=df['MarginalES'],
28 window=90).sma_indicator()
29 df[f'SMA_{180}'] = ta.trend.SMAIndicator(close=df['MarginalES'],
30 window=180).sma_indicator() # past half year trend
31
32 # Exponential Moving Average
33 df[f'EMA_{3}'] = ta.trend.EMAIndicator(close=df['MarginalES'],
34 window=3).ema_indicator()
35 df[f'EMA_{5}'] = ta.trend.EMAIndicator(close=df['MarginalES'],
36 window=5).ema_indicator()
37 df[f'EMA_{7}'] = ta.trend.EMAIndicator(close=df['MarginalES'],
38 window=7).ema_indicator()
39 df[f'EMA_{14}'] = ta.trend.EMAIndicator(close=df['MarginalES'],
40 window=14).ema_indicator()
41 df[f'EMA_{30}'] = ta.trend.EMAIndicator(close=df['MarginalES'],
42 window=30).ema_indicator()
43
44 # Momentum - Rate of Change (ROC) and RSI
```

```

33 # Rate of Change
34 df[f'ROC_{3}'] = ta.momentum.ROCIndicator(close=df['MarginalES'],
35     window=3).roc()
36 df[f'ROC_{5}'] = ta.momentum.ROCIndicator(close=df['MarginalES'],
37     window=5).roc()
38 df[f'ROC_{7}'] = ta.momentum.ROCIndicator(close=df['MarginalES'],
39     window=7).roc()
40 df[f'ROC_{14}'] = ta.momentum.ROCIndicator(close=df['MarginalES'],
41     window=14).roc()
42 df[f'ROC_{30}'] = ta.momentum.ROCIndicator(close=df['MarginalES'],
43     window=30).roc()
44
45 # Relative Strength Index
46 df[f'RSI_{5}'] = ta.momentum.RSIIndicator(close=df['MarginalES'],
47     window=5).rsi() # rapid momentum changes
48 df[f'RSI_{7}'] = ta.momentum.RSIIndicator(close=df['MarginalES'],
49     window=7).rsi()
50 df[f'RSI_{14}'] = ta.momentum.RSIIndicator(close=df['MarginalES'],
51     window=14).rsi() # biweekly momentum
52 df[f'RSI_{30}'] = ta.momentum.RSIIndicator(close=df['MarginalES'],
53     window=30).rsi()
54
55 # Volatility - Bollinger Bands Width and Standard Deviation
56 # BB Width (Bollinger Bands Width)
57 df['BB_Width_7'] = ta.volatility.BollingerBands(close=df['MarginalES'],
58     window=7, window_dev=2).bollinger_wband()
59 df['BB_Width_14'] = ta.volatility.BollingerBands(close=df['MarginalES'],
60     window=14, window_dev=2).bollinger_wband()
61
62 # Standard Deviation (STD)
63 df['STD_7'] = df['MarginalES'].rolling(window=7).std()
64 df['STD_14'] = df['MarginalES'].rolling(window=14).std()
65 df['STD_30'] = df['MarginalES'].rolling(window=30).std()
66 df['STD_90'] = df['MarginalES'].rolling(window=90).std()
67
68 # Additional contextual features
69 df['month'] = df['Datetime'].dt.month
70 df['day_of_week'] = df['Datetime'].dt.dayofweek # Monday=0
71 df['is_weekend'] = df['day_of_week'].isin([5, 6]).astype(int)
72
73 # Drop missing values generated by rolling calculations
74 df.dropna(inplace=True)
75
76 # Check for NaN or infinity values (specially in ROC)
77 print("NaN values per column:\n", df.isna().sum())
78 print("Infinity values per column:\n", df.isin([np.inf, -np.inf]).sum())
79
80 # Drop NaN or infinity values and interpolate
81 df = df.replace([np.inf, -np.inf], np.nan).dropna()
82 df = df.interpolate()

```



```
72  
73 # Save the dataset with metrics  
74 df.to_csv(f'data/ta_metrics/final_price_ta_metrics.csv', index=False  
    )
```

APPENDIX F: SIMPLE FEATURE MATRIX FUNCTION AND EXAMPLE RUN

Function:

```
1  import pandas as pd
2  import numpy as np
3
4  def create_feature_matrix(data, price_feature_window):
5      """
6          Creates a feature matrix where each row is a sliding window of
7          prices,
8          and a corresponding target vector containing the next price
9          value.
10
11         Parameters:
12         -----
13         - data: DataFrame with 'Datetime' and 'MarginalES' columns
14         - price_feature_window: Size of the prices sliding window
15
16         Returns:
17         -----
18         - X: DataFrame with sliding windows as rows
19         - y: Series with target values (next price after each window)
20         """
21         # Extract the MarginalES column
22         if 'MarginalES' in data.columns:
23             prices = data['MarginalES'].values
24         else:
25             # Assume it's the second column (index 1)
26             prices = data.iloc[:, 1].values
27
28         # Create empty matrices
29         X = np.zeros((len(prices) - price_feature_window,
30                     price_feature_window))
31         y = np.zeros(len(prices) - price_feature_window)
32
33         # Fill the matrices with the sliding windows and targets
34         for i in range(len(prices) - price_feature_window):
35             X[i, :] = prices[i:i+price_feature_window] # Window of
36                                                         prices
37             y[i] = prices[i+price_feature_window]        # Next price
38                                                         after window
39
40         # Convert to DataFrame/Series for easier use in training
41         return pd.DataFrame(X), pd.Series(y)
```

Example run:

```
1 import pandas as pd
2 from utils.matrix_builder import create_feature_matrix
3
4 # read data
5 csv_hour_file = '../data/ta_metrics/final_price_ta_metrics.csv'
6 simple_df = pd.read_csv(csv_hour_file, parse_dates=['Datetime'])
7 simple_df = simple_df[['Datetime', 'MarginalES']]
8
9 # Sliding window of 6 days
10 train_start_date = '2019-01-01'
11 train_end_date = '2019-01-07'
12
13 simple_subset = simple_df[(simple_df['Datetime'] >= train_start_date
14                             ) & (simple_df['Datetime'] <= train_end_date)]
15
16 # Number of previous days as columns (features)
17 price_feature_window = 3
18
19 X_simple, y_simple = create_feature_matrix(simple_subset,
```

APPENDIX G: EXTENDED FEATURE MATRIX FUNCTION AND EXAMPLE RUN

Function:

```
1 def create_ta_feature_matrix(dataframe, price_feature_window):
2     """
3     Creates a sliding window dataset for time series forecasting
4     where each row contains:
5     1. A window of historical prices (right-aligned)
6     2. Feature values from the most recent point in the window
7     3. Target value (next price after the window)
8
9     Parameters:
10    -----
11    dataframe : pandas.DataFrame
12        DataFrame containing at minimum 'Datetime' and 'MarginalES'
13        columns,
14        plus any additional feature columns
15    price_feature_window : int
16        Number of data points to include in each window of prices
17
18    Returns:
19    -----
20    X : pandas.DataFrame
21        Features DataFrame with:
22        - Historical prices labeled as 'price_t-n' through 'price_t-1'
23        - All additional features from the original dataframe
24    y : pandas.Series
25        Target values (price at time t)
26    """
27
28    # Input validation
29    if price_feature_window < 1:
30        raise ValueError("Feature window size must be at least 1")
31    if len(dataframe) <= price_feature_window:
32        raise ValueError(f"DataFrame must have more rows ({len(
33            dataframe)}) than price_feature_window ({
34            price_feature_window})")
35    if 'Datetime' not in dataframe.columns or 'MarginalES' not in
36        dataframe.columns:
37        raise ValueError("DataFrame must contain 'Datetime' and '
38            MarginalES' columns")
39
40    X, y = [], []
```

```

36     # Extract price data and features
37     df_prices = dataframe[['Datetime', 'MarginalES']]
38     df_features = dataframe.iloc[:, 2:] # Exclude 'Datetime' and '
        MarginalES'
39     feature_names = df_features.columns.tolist()
40
41     # Create samples from the data
42     for i in range(price_feature_window, len(df_prices)):
43         # Extract the window for prices as features (right-aligned)
44         window = df_prices.iloc[i-price_feature_window:i, 1].values.
            flatten()
45
46         # Extract corresponding feature row (from the most recent
            point in the window)
47         feature_row = df_features.iloc[i-1].values.flatten()
48
49         # Concatenate window prices with feature row
50         X.append(np.concatenate((window, feature_row)))
51         y.append(df_prices.iloc[i, 1]) # Predict current price
52
53     # Create column names for the prices feature window
54     price_columns = [f'price_t-{price_feature_window-i}' for i in
        range(price_feature_window)]
55
56     # Return DataFrame and Series with proper column names
57     X_df = pd.DataFrame(X, columns=price_columns + feature_names)
58     y_series = pd.Series(y, name='price_t')
59     return X_df, y_series

```

Example run:

```

1  import pandas as pd
2  from utils.matrix_builder import create_feature_matrix
3
4  # read data
5  csv_hour_file = '../data/ta_metrics/final_price_ta_metrics.csv'
6  features_df = pd.read_csv(csv_hour_file, parse_dates=['Datetime'])
7
8  # Sliding window of 6 days
9  train_start_date = '2019-01-01'
10 train_end_date = '2019-01-07'
11
12 feature_subset = features_df[(features_df['Datetime'] >=
    train_start_date) & (features_df['Datetime'] <= train_end_date)]
13
14 # Number of previous days as columns (features)
15 price_feature_window = 3
16
17 X_extended, y_extended = create_expanded_feature_matrix(
    feature_subset, price_feature_window)

```

APPENDIX H: EXAMPLE OF THE BASIC TRAINING PROCESS

```
1 import pandas as pd
2 from utils.matrix_builder import create_expanded_feature_matrix
3
4 # Import your preferred model here
5 from model_library import preferred_model
6
7 # Debug parameter
8 DEBUG = False
9
10 if DEBUG:
11     print("Debug mode is ON. Detailed output will be printed.")
12
13 # Load Database
14 csv_hour_file = '../data/ta_metrics/final_price_ta_metrics.csv'
15 df = pd.read_csv(csv_hour_file, parse_dates=['Datetime'])
16
17 # Define the Sliding Windows for this run
18 sliding_window = 10 # of days to train on (matrix rows)
19 price_feature_window = 3 # Window of the number of previous days as
    features (matrix columns)
20
21 # The following +1 is used to include the next row as the test set
    for model validation
22 training_sliding_window = sliding_window + 1
23
24 # Calculate number of sliding window models to train in the dataset
25 num_sliding_windows = len(df) - training_sliding_window
26 if DEBUG:
27     print(f"Number of rows in the dataset: {len(df)}")
28
29 print(f"Number of sliding windows to train: {num_sliding_windows}")
30
31 # Initialize lists to store predictions, actuals, and timestamps
32 predictions_list = []
33 actuals_list = []
34 timestamps_list = []
35
36 # Initialize the model
37 model = preferred_model
38
39 for i in range(num_sliding_windows):
40     if DEBUG:
41         print(f"Processing sliding window {i + 1}/{
            num_sliding_windows}...")
42
```

```

43     # Ensure we do not go out of bounds
44     if i + training_sliding_window >= len(df):
45         break
46
47     sliding_window_set = df.iloc[i : i + training_sliding_window]
48
49     # Create feature matrix and target variable for training
50     X_train, y_train = create_expanded_feature_matrix(
51         sliding_window_set, price_feature_window)
52
53     # Split into training matrix and prediction feature row
54     X_train_fit = X_train.iloc[:-1]
55     y_train_fit = y_train.iloc[:-1]
56
57     X_to_predict = X_train.iloc[-1:]
58     y_to_predict = y_train.iloc[-1]
59
60     # Train the model with the feature matrix
61     model.fit(X_train_fit, y_train_fit)
62
63     # Make prediction with the next row of features
64     y_predicted = model.predict(X_to_predict)
65
66     # Add bounds checking to catch extreme predictions
67     if y_predicted[0] < 0: # Avoid negative price predictions, set
68         lower limit to 0
69         y_predicted[0] = 0
70     if y_predicted[0] > 1000: # limit to 1000, avoiding unrealistic
71         predictions
72         y_predicted[0] = 1000
73
74     # Store results for this sliding window
75     predictions_list.append(y_predicted[0])
76     actuals_list.append(y_to_predict)
77     if 'Datetime' in sliding_window_set.columns:
78         timestamps_list.append(sliding_window_set.iloc[-1]['Datetime'])
79     else:
80         timestamps_list.append(i + training_sliding_window - 1)
81
82     # Create prediction vs actual DataFrame
83     prediction_df = pd.DataFrame({
84         'Timestamp': timestamps_list,
85         'Predicted': predictions_list,
86         'Actual': actuals_list
87     })

```